Smooth Particle Open-source in-Operation Fuel Fragmentation (SPOOFF) tool

Installation Guide and User documentation

Created by the United Kingdom National Nuclear Laboratory (UKNNL) in partnership with the University of Bangor Nuclear Futures Centre under the OperaHPC project

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1 Overview

The Smoothed Particle Open-source in-Operation Fuel Fragmentation (SPOOFF) tool is a prototype SPH style simulation code created through modification of the MACHDYN [1] package within LAMMPS [2]. This means that the tool has MPI implementation for large scale parallel HPC use cases. The method represents materials as a continuum with mesh free interpolation points where continuum quantities are calculated at any point based on weighted sums of the interpolation point property values [REF SPH]. The method allows fragmentation through a total-Lagrangian formulation[REF tlsph] where pairs of interacting interpolation points are only considered based on the initial reference configuration and are removed as interacting following failure.

The tool has been developed by UKNNL in association with Bangor university for the OperaHPC project [3]. As such the tool remains a work in progress prototype with multiple known issues and missing models which are planned for future versions.

This documentation details modifications made that cause it to differ from the implementation of the MACHDYN tool it is built on top of. As such it is important that the reader familiarise themselves with the MACHDYN documentation as well as this documentation.

Expected results for all the verification cases are included within this document as only the input files are included within the GitHub repository for each due to size restrictions. For each case a brief description is given and the expected results shown.

2 Installation

Unfortunately at this stage installation is not straight forward however a later goal is to include install scripts to streamline the process. SPOOFF is not included in LAMMPS as standard and therefore is not updated with versions of LAMMPS.

The current version of SPOOFF was developed on top of LAMMPS version 02Aug2023 is currently incompatible with any other version. The modifications we made to the existing MACHDYN package within LAMMPS however they have been separated into there own package to allow for MACHDYN to still be installed along side if the user desires.

There exists 2 routes to building LAMMPS, traditional make and cmake. In the latest version of LAMMPS MACHDYN has moved to allow builds only with cmake however this documentation presents the route to install through the traditional make process. It is the intention to add documentation for the cmake process at a later date.

Because the traditional make process is used the user will require a Linux environment. For this example Ubuntu is used with the windows subsystem for Linux. As well as Ubuntu it is advised that the user download the free Ovito software to allow quick and easy post processing of results.

2.1 Install steps (Windows using Ubuntu):

1. Install Ubuntu (Be sure to enable WSL)
   https://ubuntu.com/download
2. Install Ovito (not necessary to run but advised to process results)
   https://www.ovito.org/
3. Download Lammps 02Aug2023 tar from
   https://github.com/lammps/lammps/releases/tag/stable_2Aug2023_update4
4. Unpack tar using Ubuntu command
   ```
   tar -xzvf lammps*.tar.gz
   ```
5. Download SPOOFF
   https://github.com/mhorton7882/SPOOFF/tree/main
6. Copy contents of SPOOF_Main /lib/spooff into lammps-2Aug2023\lib\spooff
7. Copy contents of SPOOF_Main/src into lammps-2Aug2023\src
   (This will overwrite the atom.cpp and atom.h files and create a new package folder SPOOFF)
8. Using Ubuntu enter the folder lammps-2Aug2023\lib\spooff and download and unpack eigen3.4.0 with commands:
   ```
   wget https://gitlab.com/libeigen/eigen/-/archive/3.4.0/eigen-3.4.0.tar.gz
   tar -xf eigen-3.4.0.tar.gz
   ```

9. While in folder lammps-2Aug2023\lib\spooff with Ubuntu create a link to eigen-3.4.0 with command:
   ```
   ln -s ./eigen-3.4.0 includelink
   ```
10. Using Ubuntu enter the folder lammps-2Aug2023\src ready to compile the code
11. If you do not already have it obtain the openmpi compiler:
    ```
    sudo apt update
    sudo apt install openmpi-bin openmpi-common libopenmpi-dev
    ```
12. If you do not already have it obtain the g++ compiler:
    ```
    sudo apt-get update
    sudo apt-get install g++
    ```
13. Now within the folder lammps-2Aug2023\src compile the code with 3 commands (the full compile takes a long time):
    ```
    make yes-sph
    make yes-spooff
    make mpi
    ```

3 Code Modifications and Commands

The Package is a modified version of the MACHDYN package, therefore it is essential that the user is familiar with the MACHDYN user documentation [1] before reading through the modifications and this document will seek not duplicate information but highlight changes and additions.

In general the rule is that wherever a command previously used "smd" it is replaced with "spoof" the only exceptions are for the set commands for the mass density and contact radius which are still written as:

```
set group all smd/contact/radius value
set group all smd/mass/density value
```

3.1 Pair Styles Syntax

The package is structures such that most of the core physics is defined through the pair styles however this package heavily utilises fix commands to change property values to allow for correlations to be used. The pair styles tlsph and hertz have been modified from the MACHDYN implementation and these modifications are discussed below. A new thermal pair style has also been added. The ulsph pair style from the MACHDYN package exists within spoof but is unmodified and un tested and therefore may have issues or conflicts. This is left in as future development will modify this pair style to allow for directly modelling the gas phase in the gap.

3.1.1 Tlsph

The tlsph pair style has been modified to include heat flow, thermal expansion, principal stress and strain failure conditions and per particle temperature dependant properties. Each of these modifications is discussed in further depth in the section that follow. Here we highlight changes to the syntax for calling the pair style:

```
pair_style spooff/tlsph
pair_coeff 1 1 *COMMON density youngs poissons artificial1 artificial2
hourglass specific_heat conductivity heat_generation &
*STRENGTH_LINEAR &
*EOS_THERMAL reference_temp linear_expansion &
*FAILURE_MAX_PRINCIPAL_STRESS failure_strain &
*END
```

Name	Expected Value
"density"	The material density (should be consitent with the values set for particle volume and mass density)
"youngs"	The material elastic modulus– this can be over written with a fix

"poissons"	The material Poisson's ratio– this can be over written with a fix
"artificial1"	The linear artificial viscosity coefficient (between 0-1)
"artificial2"	The quadratic artificial viscosity coefficient (between 0-1)
"hourglass"	The hourglass coefficient (set to 0 throughout simulations run for this work- see MACHDYN user manual [1])
"specific_heat"	The material specific heat – this can be over written with a fix
"conductivity"	The material thermal conductivity – this can be over written with a fix
"heat_generation"	The material heat generation per unit mass – this can be over written with a fix
"reference_temp"	The material reference temperature where it is at rest (this is usually the initial set staring temperature for the simulation) – this can be over written with a fix
"linear_expansion"	The material linear expansion coefficient– this can be over written with a fix
"failure_strain"	The material failure strain (converted to stress for this failure type -see – this can be over written with a fix

3.1.2 Hertz

Contact is modelled through a pairwise hertz contact force assuming contact between two regular spheres as in the MACHDYN package [1]. For this application two modification have been made to the pair style. Force is now considered with a relative elastic modulus a viscous damping force has been added to prevent unstable energy buildups such that the force is now calculated as:

$$F_{ij} = S_F \left[\frac{\sqrt{R_{ij}}}{\frac{1-\nu_i^2}{E_i} + \frac{1-\nu_j^2}{E_j}} \delta_{ij}^{\frac{3}{2}} + \frac{S_D 0.587(1-c_r)\sqrt{R_{ij}}}{\frac{1-\nu_i^2}{E_i} + \frac{1-\nu_j^2}{E_j}} \delta_{ij}^{\frac{1}{2}} v_{ij} \right] \quad (1)$$

Where R is the geometric mean of the particle contact radii, δ is the over lap of contact radii (the penetration depth), v is the relative velocity of the collision, E is the elastic modulus, ν is Poisson's ratio and c_r is the coefficient of restitution. The viscous damping is based on [4] and is selected as collisions are assumed to be low speed. User scaling factors for the total force S_F and damping S_D have been added to allow the forces to be further adjusted.

```
pair_style hybrid/overlay spooff/tlsph spooff/hertz distance_scale
pair_coeff 1 1 spooff/tlsph *COMMON density youngs poissons artificial1
artificial2 hourglass specific_heat conductivity heat_generation &
*STRENGTH_LINEAR &
*EOS_THERMAL reference_temp linear_expansion &
*FAILURE_MAX_PRINCIPAL_STRESS failure_strain &
*END
pair_coeff 1 1 spooff/hertz force_scale damping_scale restitution
```

Name	Expected Value
"distance_scale"	The is how the contact radius is scaled to determine when the interaction starts.
"force_scale"	This is the factor S_F from equation (1) A typical value is between 0.01 and 0.5.
"damping_scale"	This is the factor S_D from equation (1). A typical values is 1.
"restitution"	This is the coefficient of restitution for the damping force and should be between 0 and 1 where 1 is a perfect elastic collision.

It should be noted that self contact is supported however the contact radius is scaled to half for points that are already interacting within the reference configuration so that they only push apart once they have moved too close together – indicating failure.

High speed collisions are numerically unstable for the method and therefore velocity limiting can be used to slow the movement of fragmented material as the discretisation scheme is known to have issues with the dissipation of energy and this can help address it[REF]. To do this velocity limiting can be added to the tlsph integration fix as:

```
fix fixname all spoof/tlsph/intergrate limit_velocity value
```

Where "value" is the maximum allowed velocity at each step. Note this can break conservation of energy.

3.1.3 Thermal

As discussed in the tlsph pair style, the loss of particle pairs under damage causes issues with temperature across the material as heat no longer propagates between the pair. To address this potential issue an alternative pair style approach to temperature calculation is included. To use this the key word *THERMAL_OFF should be added to the tlsph pair style and the alternative thermal style overlayed onto the tlsph style. This style is not intended to be used in isolation as the tlsph pair style still initialises the values for the material. An example implementation is shown here:

```
pair_style hybrid/overlay spooff/tlsph spooff/thermal
pair_coeff 1 1 spooff/tlsph *COMMON density youngs poissons artificial1
artificial2 hourglass specific_heat conductivity heat_generation &
*STRENGTH_LINEAR &
*EOS_THERMAL reference_temp linear_expansion &
*FAILURE_MAX_PRINCIPAL_STRESS failure_strain &
*THERMAL_OFF &
*END
pair_coeff 1 1 spooff/thermal
```

3.2 Fix Syntax

This package heavily utilises fix commands to change property values to allow for correlations to be used, for only partial regions to be set some property values and for property values to be changed as the simulation progresses.

The majority of fix command implemented use the same general format for consistency. An example is shown below:

```
fix fixname groupname spooff/setpropertyname scale mode
```

The “fixname” is an identifier set by the user to allow unfix to be specified at a later date.

The “groupname” specifies the region of particles on which the fix acts

The choice of fix is usually labelled with the format “spoof/setpropertyname” where property name is specific to what is being set.

The values “scale” and “mode” define the values to be set. Generally within this package the modes are 0 (constant value), 1 (UO_2 correlation) and 2 (Zircalloy correlation). When “mode” = 0 is used then the constant value set is given by “scale”. When “mode” = 1 or 2 is used then the value “scale” is multiplicative with the correlation calculation to allow the user to change the base units of the simulation.

3.3 Heat Flow

Thermal evolution is included within the code through the following SPH discretised equation:

$$\dot{e} = \frac{1}{\rho_i}\sum_j \frac{m_i m_j}{m_i+m_j}\frac{\rho_i+\rho_j}{\rho_i\rho_j}\frac{2}{\frac{1}{\kappa_i}+\frac{1}{\kappa_j}}\frac{c_{p_i}+c_{p_j}}{c_{p_i}c_{p_j}}\left(e_i-e_j\right)\nabla_i W\left(\left|r_i-r_j\right|,h\right) + \dot{q}_i m_i \qquad (2)$$

Where e is the internal thermal energy, ρ the density, m the mass, κ the thermal conductivity, c_p the specific heat, r the position and $\dot{q}$ the heat generation per unit mass for interpolation points described by the index i, j. h represents the cutoff distance of the kernel function W.

Heat flow can be included utilising either the tlsph pair style or the thermal pair style (see above for syntax and understanding of the difference between the methods).

3.3.1 Heat Generation

Heat generation is considered per unit mass as seen in equation (*2*).

A fix exists to allow the user to define heat generation for only a region of the simulation and change it at different simulation intervals. This changes the particle values of $\dot{q}_i$ and has base units of W/kg.

```
fix fixname groupname spooff/setheatgen scale mode
```

3.3.2 Thermal Equation of State

Thermal expansion is added through a temperature dependant equation of state as:

$$P_i = -B_i \frac{\rho_i\left(1 + 3\alpha(T_i - T_0)\right)}{\rho_0} \tag{3}$$

This is a modified version of the linear equation of state implemented within MACHDYN.

The equation of state is evoked for the tlsph pair style using the following key work syntax:

```
*EOS_THERMAL T0 alpha &
```

T0 defines the reference temperature for the rest state of the material and alpha is the linear thermal expansion coefficient for the material. Both of these properties can be reassigned through fixes. The linear expansion fix is given in the thermal properties section with correlations. The T0 fix is used as:

```
fix fixname groupname spooff/setreferencetemperature scale mode
```

The only mode currently implemented for this is mode 0 (constant) however more could be added by the user if required.

3.3.3 Temperature and Energy

The temperature is stored per atom now as well as the internal energy. The initial temperature is calculated through the values supplied to the tlsph pair style for specific heat and density as well as the user set internal energy value set as:

```
set group groupname sph/e value
```

From this the initial temperature is calculated as:

$$T_i = \frac{e_i}{\rho_i V_i c_{p_i}} \tag{4}$$

Where V is the volume set by the user. From this point the temperature evolution is time integrated along with the internal energy evolution. This is done to bypass potential issues with any temperature dependency of the specific heat.

Temperature units are in K. The temperature and internal energy can be computed for output as:

```
compute e all spooff/internal/energy
compute t all spooff/temperature
dump dump_id all custom 1000 dump.LAMMPS id type x y z vx vy vz &
c_e c_t
dump_modify dump_id first yes
```

The work done calculated by MACHDYN was previously also assigned to the internal energy value however this causes issues as not all work done arises as temperature due to elastic potential energy etc. Because of this a new variable is created and time integrated to track the work done energy separately. This value can also be computed by the command:

```
compute ID all spooff/potential/energy
```

The temperature can also be fixed in a region with the syntax:

```
fix fixname groupname spooff/settemp fixrate
```

The value of “fixrate” sets the rate of change of the temperature so set to 0 for constant, unchanging temperature.

3.3.4 Heat flux

The heat flux vector can now be calculated through a compute:

```
compute Q all smd/heatflux
dump dump_id all custom 1000 dump.LAMMPS id type x y z vx vy vz &
c_Q[1] c_Q[2] c_Q[3]
dump_modify dump_id first yes
```

If the tlsph pair style is used then the gradient correction is included in the calculation however this is not currently calculated for the thermal pair style. This is left to a future implementation to be aligned.

3.3.5 Convective boundary

To create a convective boundary region it must be a set of tlsph particle on the exterior of the system. The convective command for the particle group uses the group name twice in the sytax:

```
fix fixname groupname spooff/setconvection hvalue Tinf region groupname
```

The hvalue is the coefficient of convection and the Tinf is the temperature of the coolant at an infinite distance from the boundary.

There are some issues known with this fix. The specific heat must be given a constant value for the boundary particles to avoid numerical instability. The method is also sensitive to the particle spacing with more work underway to improve the method for future use. For now, as a temporary measure, the hvalue can be adjusted if the particle spacing shifts the expected result.

The convective boundary condition is enforced by reassigning boundary interpolation point temperatures based on the approximation of newts law of cooling:

$$T_i = \frac{T_B + T_\infty H|r_i - r_B|/\kappa_i}{1 + H|r_i - r_B|/\kappa_i} \tag{5}$$

Here the subscript B refers to the nearest system particle to the boundary. H defines the coefficient of convection and T_∞ defines the temperature of the convective coolant. This conditions is based on previous work [5].

It should be noted this fix loops over all pairs in the reference configuration and so has a computational cost associated with it for very large simulations.

3.4 Thermal Property Correlations

All thermal property correlations are taken from the MATPRO V11 [6] documentation and are implemented through fixes. The correlations are given here for reference. Some properties are dependant on the porosity (or adjusted density). For these the porosity can be set using fix:

```
fix fixname groupname spooff/setporosity scale mode
```

Where the convention is taken that the default porosity value =1if not set by the user and this is equivalent to no pores present or an adjusted density the same as that set.

Where oxygen to metal ratios are present in MATPRO correlations for UO_2 these are just assumed to be a perfect ratio of 2 for now with no ability for the user to adjust (this is left to future development).

3.4.1 Thermal Conductivity

The thermal conductivity is set with fix:

```
fix fixname groupname spooff/setconductivity scale mode
```

The per particle conductivity can be calculated by the compute command:

```
compute ID all spooff/conductivity
```

Mode 0 gives assigns the scale value

Mode 1 is the UO_2 correlation for $T \leq 1650°C$ defined by:

$$\kappa_i = scale * P_{c_i}\left[\frac{40.4}{464 + T_i} + 1.216\text{x}10^{-4}\exp(0.0191T_i)\right] \quad (6)$$

And for $T > 1650°C$ defined by:

$$\kappa_i = scale * P_{c_i}[5 + 1.216\text{x}10^{-4}\exp(0.0191T_i)] \quad (7)$$

Where P_c is the porosity correction factor.

Mode 2 is the Zircalloy correlation defined by:

$$\kappa_i = scale * \left[7.51 + 2.09\text{x}10^{-2}T_i - 1.45\text{x}10^{-5}T_i^2 + 7.67\text{x}10^{-9}T_i^3\right] \quad (8)$$

The units are W/mK

3.4.2 Specific Heat

The thermal conductivity is set with fix:

```
fix fixname groupname spooff/setspecifictheat scale mode
```

The per particle specific heat can be calculated by the compute command:

```
compute ID all spooff/specificheat
```

Mode 0 gives assigns the scale value

Mode 1 is the UO_2 correlation defined by:

$$c_{p_i} = scale * \frac{296.7\,(535.285)^2 \exp\left(\frac{535.285}{T_i}\right)}{T_i^2\left[exp\left(\frac{535.285}{T_i}\right) - 1\right]^2} + 0.0243T_i + \frac{1.38\text{x}10^{13}}{8.31T_i^2}\exp\left(-\frac{1.577\text{x}10^5}{8.31T_i}\right) \quad (9)$$

Mode 2 is the Zircalloy correlation. This created as a polynomial fit to the MATPRO data and is valid between $300K < T < 1090K$ (above and below it is just assumed constant):

$$c_{p_i} = scale * [164.28 + 0.5581T_i - 6.504\text{x}10^{-4}T_i^2 + 2.896\text{x}10^{-7}T_i^3] \quad (10)$$

The units are J/kgK

3.4.3 Thermal Expansion

The linear thermal expansion is set with fix:

```
fix fixname groupname spooff/setlinearexpansion scale mode
```

The per particle linear thermal expansion can be calculated by the compute command:

```
compute ID all spooff/linear_expansion
```

Mode 0 gives assigns the scale value.

Mode 1 is the UO_2 correlation for defined by:

$$\alpha_i = scale * [-4.972\text{x}10^{-4} + 7.107\text{x}10^{-6}(T_i - T_0) + 2.581\text{x}10^{-9}(T_i^2 - T_0^2) + 1.14\text{x}10^{-13}(T_i^3 - T_0^3)]/(T_i - T_0) \quad (11)$$

This is only valid up to the melting temperature however melting behaviour is not yet implemented. The calculation converts the extension to a coefficient α to align with the constant value input method. The values is unitless but the scale factor is left in for consistency of definitions.

Mode 2 is the Zircalloy correlation (assumed to be the same in all directions and taken as the axial extension). For $T \leq 1073K$ defined by:

$$\alpha_i = scale * [4.44\text{x}10^{-6}(T_i - T_0) - 1.24\text{x}10^{-3}]/(T_i - T_0) \quad (12)$$

And for $T > 1073K$ defined by:

$$\alpha_i = scale * [9.7\text{x}10^{-6}(T_i - T_0) - 1.1\text{x}10^{-2}]/(T_i - T_0) \quad (13)$$

3.5 Failure Models

All but the pairwise strain failure model uses the specified dame criteria to determine the onset of damage. Once damage has occurred the point wise failure mechanism is used to degrade and remove pair interactions:

$$D_{ij} = \frac{|r_i - r_j| - |r_i^f - r_j^f|}{2|r_i^f - r_j^f|} \tag{14}$$

Where D is the degradation of bond ij which is assumed to failure completely when the pair strain is more than twice what it was at the point of failure (denoted by superscript f). The bond degradation is used to scale the contribution of interaction between the pair until no interaction is considered once $D > 1$.

One limitation of considering the thermal expansion is that pairwise strains are far larger than the expected failure strains considering only the elastic regime relation and therefore a principal stress failure condition must be used.

3.5.1 Principal Stress

The principal stress failure criteria checks if any of the eigen values of the stress tensor are greater than the specified failure stress. However because all other failure mechanisms are determined by strain not stress values the decision is made to implement the principal stress failure with strain values as well to allow for consistency with the failure fix method discussed below. Therefore the failure strains to failure stress relation is always assumed based on the elastic regime using Hooke's law:

$$\sigma_i^f = E_i \epsilon_i^f \tag{15}$$

Where E is young's modulus, ϵ^f is the failure strain and σ^f is the failure stress. Remember that because of the thermal expansion this relationship will not be true at failure of the pellet so it matters weather principal stress of principal strain are used. The principal stress failure model is activated within the tlsph pair style through the addition of the following key work:

```
*FAILURE_MAX_PRINCIPAL_STRESS Failure_Strain_Value &
```

3.5.2 Principal Strain

The principal strain is added in line with the principal stress failure. The condition checks to see if any of the eigen values of the strain tensor are higher than the supplied value and uses this to flag the onset of damage. The key word implementation is:

```
*FAILURE_MAX_PRINCIPAL_STRAIN Failure_Strain_Value &
```

3.6 Mechanical Property Correlations

Where available the mechanical property correlations are taken from the MATPRO V11 [6] documentation unless otherwise stated and are implemented through fixes. The correlations are given here for reference.

3.6.1 Young's Modulus

Temperature dependant young's modulus is included The fix has the form:

```
fix fixname groupname spooff/setyoungs scale mode
```

Mode 0 gives assigns the scale value as a constant.

Mode 1 is the UO_2 correlation for young's modulus defined by (Valid up to melting):

$$E_i = scale * [2.334\text{x}10^{11}(1 - 2.752(1 - P_c))(1 - 1.0915\text{x}10^{-4}T_i] \quad (16)$$

Mode 2 is the Zircalloy correlation for young's modulus defined for $T < 1173K$ (Alpha phase with neutron fluence, oxidation and cold work ignored) as:

$$E_i = scale * (1.088\text{x}10^{11} - 5.475\text{x}10^{7}T_i) \quad (17)$$

For the Zircalloy beta phase (assumed $T > 1173K$):

$$E_i = scale * (9.21\text{x}10^{10} - 4.05\text{x}10^{7}T_i) \quad (18)$$

A compute is also included:

```
compute ID all spooff/youngs
```

3.6.2 Poisson's Ratio

This has a fix however currently only mode 0 for constant value is implemented:

```
fix fixname groupname spooff/setpoissons scale mode
```

A compute is also included:

```
compute ID all spooff/poissons
```

3.6.3 Yield Stress

Yield stress correlations are added for use with the "STRENGTH_LINEAR_PLASTIC" model. The fix has the form:

```
fix fixname groupname spooff/setyield scale mode shape minimum
```

Mode 0 gives assigns the scale value.

Mode 1 is the UO_2 correlation for fracture stress (converted to strain) defined by:

$$\epsilon_i^f = scale * \tag{19}$$

Mode 2 is the Zircalloy correlation for fracture stress (converted to strain) defined by:

$$\epsilon_i^f = scale * \tag{20}$$

A compute is also included:

```
compute ID all spooff/yield
```

3.6.4 Failure Fix

To allow relation of the material to occur a fix is added for the failure strain (which is converted to stress using Hooke's law if required). The fix has the form:

```
fix fixname groupname spooff/setfailure scale mode shape minimum
```

Mode 0 gives assigns the scale value.

Mode 1 is the UO_2 correlation for fracture stress (converted to strain) defined by:

$$\epsilon_i^f = scale * \frac{\left[1.7\text{x}10^8\sqrt{1-2.62(1-P_c)}\exp\left(-\frac{1590}{8.314T_i}\right)\right]}{E_i} \tag{21}$$

This is up to $T = 1000K$ beyond which it is assumed constant.

Mode 2 is the Zircalloy correlation for fracture stress (converted to strain) defined by:

$$\epsilon_i^f = scale * \tag{22}$$

Mode 3 is a Weibull distribution and utilises the extra two values of shape m and minimum ϵ_{min} . The mode also uses the scale value as the characteristic value ϵ_0. To

calculate this a random number R_N is calculated between 0 and 1. The strain failure value is then calculated as:

$$\epsilon_i^f = \epsilon_0[-\ln(1 - R_N)\,]^m + \epsilon_{min} \tag{23}$$

NOTE: all 4 values must be supplied for all modes however the shape and minimum values will not have any effect unless the mode is set to 3.

A compute is also included:

```
compute ID all spooff/failure
```

3.7 Damage Tracking

Two types of compute are added for tracking the material damage. Per particle is the standard type of compute and can be used with the standard dump file. Per "bond" is a local compute that outputs information about each pair i, j in the reference configuration and requires a separate dump local file

3.7.1 Per Particle

The damage compute calculates 4 different values to assess the per particle damage. The relevance of each metric depends on the failure mode chosen, not all values will be relevant for all failure types and some values are included to support troubleshooting. The 4 values are:

1. The ratio of current neighbours n to initial neighbours n_0 as $d_1 = 1 - (n/n_0)$. This is the most useful metric and is relevant for all damage types. A value 0 indicates no damage, a value 1 indicates completely damaged (no neighbours).
2. The average degradation of pair interactions for the particle:

$$d_2 = \frac{\sum_{j\epsilon n_0} D_{ij}}{n_0} \tag{24}$$

3. The average "bond energy" for the remaining bonds:

$$d_3 = \frac{\sum_{j\epsilon n}\left|r_i^f - r_j^f\right|}{n} \tag{25}$$

4. The associated damage value (this is most often a flag (0 or 1) to indicate the onset of damage where 0 is not damage and 1 is damage has begun.

An example use of the compute is below:

```
compute dam all spooff/damage
dump dump_id all custom 1000 dump.LAMMPS id type x y z vx vy vz &
c_dam[1] c_dam[2] c_dam[3] c_dam[4]
dump_modify dump_id first yes
```

It should be noted this compute loops over all pairs in the reference configuration and so has a computational cost associated with it for very large simulations. Future work may separate the 4 variables to avoid this for some cases.

3.7.2 Per Bond

The local damage compute offers 3 per pair values:

Damage is the degradation value but limited to a highest value of 1 ($MIN[1, D_{ij}]$). Although failure occurs at 1 it is possible for the degradation to be given a higher value in the instantaneous step where the failure is calculated. Limiting this to 1 is useful for visualising outputs.

Degradation is simply the value D_{ij}.

Energy is the max strain when the onset of damage occurs $|r_i^f - r_j^f|$. Again it is possible for this to be larger than the pairwise strain failure values (if utilised).

An example use of the compute is below:

```
compute pair all damage/local patom1 patom2 damage degredation energy
dump dumplocal all local 1000 dumplocal.LAMMPS c_pair[*]
dump_modify  dumplocal colname 1 patom1 colname 2 patom2 colname 3
damage colname 4 degredation colname 5 energy
```

It should be noted this compute loops over all pairs in the reference configuration and so has a computational cost associated with it for very large simulations.

3.8 Cylindrical Co-ordinate System

Cylindrical co-ordinates are far more helpful for fuel pellet analysis therefore computes exist for the co-ordinates, stress and strain tensors and heat flux vector to be output in cylindrical co-ordinates. An example use is shown here:

```
compute r all spooff/cylindrical
compute S all spooff/tlsph/stress_cylindrical
compute E all spooff/tlsph/strain_cylindrical
compute Q all smd/heatflux_cylindrical
dump dump_id all custom 1000 dump.LAMMPS id type x y z vx vy vz &
c_r[1] c_r[2] c_r[3] &
c_S[1] c_S[2] c_S[4] &
c_E[1] c_E[2] c_E[4] &
c_Q[1] c_Q[2] c_Q[3]
dump_modify dump_id first yes
```

4 Example Simulation

Steps to run a simulation are given as well as an example input script. It is advised that simulations are run from a folder within the lammps folder structure however if this is not the case then the path to the generated build executable will need updating. Steps to run a simulation are given below:

1. Ensure you have followed the installation steps and have an executable file "lmp_mpi" in your src folder
2. Create a folder within the lammps folder space "lammps-2Aug2023\Verification Cases\Test" (or copy in the existing verification cases downloaded with the source code)
3. Within the folder "Test" create a text file that will be used as the input script and name it "run.lmp" – note .lmp is a common extension used for lamps inputs but not strictly necessary.
4. Populate the "run.lmp" file with the input commands
5. Now from the Ubuntu console navigate to the folder "lammps-2Aug2023\Verification Cases\Test". From here you can execute the run.lmp file with the command:
   `mpirun -np 4 ../../src/lmp_mpi -in 'run.lmp'`

In general you must run lamps from the directory where you input script is located however you can replace the path to lmp_mpi to adjust where the input folder is located relative to the executable. The 4 in the above command gives the number of processors on which to run.

An example input script is given in section 4.1 and each section is broken down in section 4.2.

4.1 Example run.lmp file

```
##########################################################################
# 1.MATERIAL PARAMETERS   units in grams and millimeters and seconds
##########################################################################
variable E equal 2.3E11 # Young's modulus   g/mms2
variable nu equal 0.314 # Poisson ratio
variable rho equal 1.1E-2 # initial mass density   g/mm3
variable q1 equal 1 # standard artificial viscosity linear coefficient
variable q2 equal 0 # standard artificial viscosity quadratic coefficient
variable hg equal 0 # hourglass control coefficient
variable cp equal 3E8 # heat capacity of material   mm2/s3K
variable k equal 3E6 # Thermal Conductivity of material   gmm/s3K
variable alpha equal 0 # thermal expansion of material
variable T0 equal 500 # rest temperature of material   K
variable EF equal 5E3 # strain failure – far higher than the actual value
variable HM equal 0 # heat per unit mass   mm2/s3

##########################################################################
# 2.INITIALIZE LAMMPS
##########################################################################
dimension 2
```

```
units si
boundary sm sm p # simulation box boundaries
atom_style spooff
atom_modify map array
comm_modify vel yes
neigh_modify every 10 delay 0 check yes # re-build neighbor list every 10 steps
newton off

##########################################################################
# 3.CREATE INITIAL GEOMETRY
##########################################################################
variable l0 equal 0.2 # lattice spacing for creating particles
lattice sq ${l0}
region box block -6 6 -6 6 -1 1 units box
create_box 1 box
region cyl cylinder z 0 0 6 -0 0.05 units box
create_atoms 1 region cyl
region inner cylinder z 0 0 4.1 -0.1 0.1 units box
region clad_in cylinder z 0 0 4.1 -0.1 0.1 units box side out
region clad_out cylinder z 0 0 5 -0.1 0.1 units box
region cyl_in cylinder z 0 0 5 -0.1 0.1 units box side out
region cyl_out cylinder z 0 0 6 -0.1 0.1 units box
region outer intersect 2 cyl_in cyl_out
region main intersect 2 clad_in clad_out
group fuel region inner
group cladding region main
group outer region outer
group tlsph type 1

###########################################################################
# 4.DISCRETIZATION PARAMETERS AND PROPERTY VALUES
###########################################################################
variable h equal 3.01*${l0} # SPH smoothing kernel radius
variable skin equal ${h} # Verlet list range
set group all diameter ${h} # set SPH kernel radius
neighbor ${skin} bin
variable vol_one equal ${l0}^2 # volume of one particle -- assuming unit
thickness
set group all volume ${vol_one}
set group all smd/mass/density ${rho}
variable energy equal 300*(${rho}*${vol_one}*${cp})
set group all sph/e ${energy}

fix convec_fix outer spooff/setconvection 3E7 300 region outer
fix k_fuel_fix fuel spooff/setconductivity 1E6 1
fix cp_fuel_fix fuel spooff/setspecificheat 1E6 1
fix k_cladding_fix cladding spooff/setconductivity 1E6 2
fix cp_cladding_fix cladding spooff/setspecificheat 1E6 2
fix Hg_fuel fuel spooff/setheatgen 0.8E11 0

###########################################################################
# 5.INTERACTION PHYSICS / MATERIAL MODEL
###########################################################################
pair_style spooff/tlsph
pair_coeff 1 1 *COMMON ${rho} ${E} ${nu} ${q1} ${q2} ${hg} ${cp} ${k} ${HM} &
*STRENGTH_LINEAR &
*EOS_THERMAL ${T0} ${alpha} &
*FAILURE_MAX_PAIRWISE_STRAIN ${EF} &
*END
```

```
##############################################################################
# 6.TIME INTEGRATION
##############################################################################
fix integration_fix tlsph spooff/integrate_tlsph
##############################################################################
# 7.SPECIFY TRAJECTORY OUTPUT
##############################################################################
compute e all spooff/internal/energy
compute t all spooff/temperature
compute Q all spooff/heatflux_cylindrical
compute r all spooff/cylindrical
dump dump_id all custom 100 dump.LAMMPS id type x y z vx vy vz &
c_e c_t &
c_Q[1] c_Q[2] c_Q[3] &
c_r[1] c_r[2] c_r[3]
dump_modify dump_id first yes
##############################################################################
# 8.RUN SIMULATION
##############################################################################
timestep 0.001    # fixed this based on time step limit calc for thermal sim.
run 50000
```

4.2 Structure of input file

Here we discuss each numbered section of the above script and its meaning:

1. The material parameters are stored as names variable here ready to be passed to the pair style and set command latter- this makes the simulation script more readable. The convection for this simulation is to work in units of grams and millimetres however this is not a requirement and very small or large numbers can be used instead.
2. These commands set up the Lammps simulation. In general the only command that are lickley to be changed are to move from a 2D to a 3D simulation. 2D simulations must be periodic defined by line "`boundary sm sm p`" for which the p marks the z direction as periodic. It is important for 2D spoof simulations that the effect of potential periodic interactions and particle volumes are correctly considered based on the chosen kernel length to avoid errors.
3. The initial geometry section sets up a simulation box then sets up atoms within that box and defines different regions which are then used to create grouped names of particle for use with future fixes. The geometry being set up by this section is illustrated in Figure 1. All particles which use the tlsph pair type must be assigned to a group called tlsph.

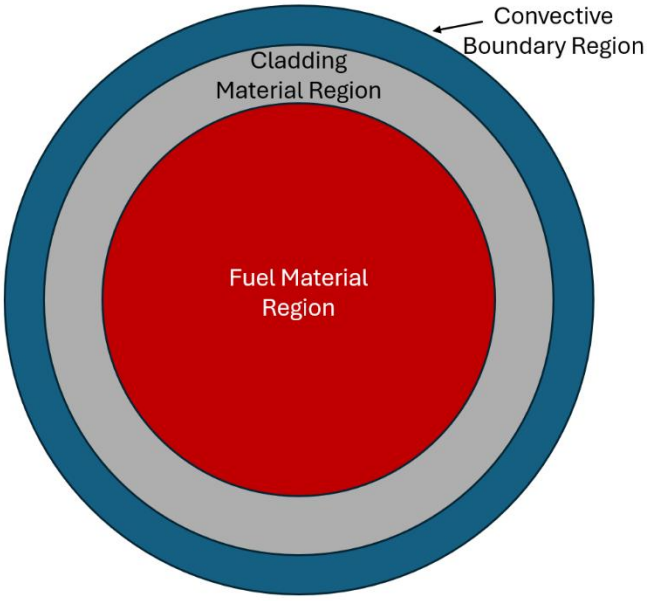

Figure 1: Diagram of the conceptual model used for the input script given in section4.1.

4. Discretization parameters include the kernel length which is set through the skin parameter. The volume is assumed unit thickness – for this to be valid the box size in the z direction has been set larger than the interaction distance of the kernel. The initial temperature has been set to 300K through setting the energy.
   The outer region is set to have a temperature profile defined by convective boundary conditions. The property values for specific heat and thermal conductivity are set to the property correlations for the fuel and cladding regions and the scale factor is included to convert to the grams millimetres units system used. The heat generation is also set in the fuel region. These properties overwrite the general constant values which are passed to the pair style later.
5. The pair style defines the physics chosen. In this case the system is at rest as thermal expansion is set to 0 so the strength linear model is not actually relevant nor are the mechanical property values. Some of the thermal property values defined in this pair style are overwritten by the fixes implemented in section 4.
6. The time integration is imposed by this fix command.
7. This section writes an output file “dump.LAMMPS” with the chosen computes and co-ordinate and velocity values. This file type is read easily in the Ovito [7] software. The number 100 states how many time steps pass between each output write.
8. Finally the time step is defined and the simulation is executed with the “run” command with the number of steps to calculate give (50000 here). Sections 4,5,6,7 can be added in any order and should give the same results.

5 Verification Cases

For each verification case:

- The input script location is given
- A brief description of the case is given
- The analytical solution is given
- The expected results are shown
- The expected convergence is shown

To recreate the cases each has a run.lmp file associated with it. There are generally multiple input files for different particle spacings (dp) each contained in its own folder to allow for the order of convergence of the case to be calculated.

Each case has a post processing R script associated with it which is capable of reproducing the graphs shown. Note that the R script will have paths that need overwriting.

5.1 Thermal

5.1.1 1D Heat flow

Found under: Verification Cases/1D Transient Heatflow

The case assumes a 1D material with length $L = 20$mm and constant temperature zones at both ends ($x < -10, x > 10$) which are set to a temperature $T = 0K$. The initial system temperature is set to $T_0 = 1K$. The analytical solution for the evolution of temperature with time is given by:

$$T(x,t) = \frac{4T_0}{\pi}\sum_m \frac{-1^{m-1}}{2m-1}\cos\left(\frac{(2m-1)\pi x}{L}\right)e^{-\left(\frac{(2m-1)^2\pi^2\kappa t}{\rho c_p L^2}\right)} \tag{26}$$

Where κ is the thermal conductivity, c_p is the specific heat and ρ is the density.

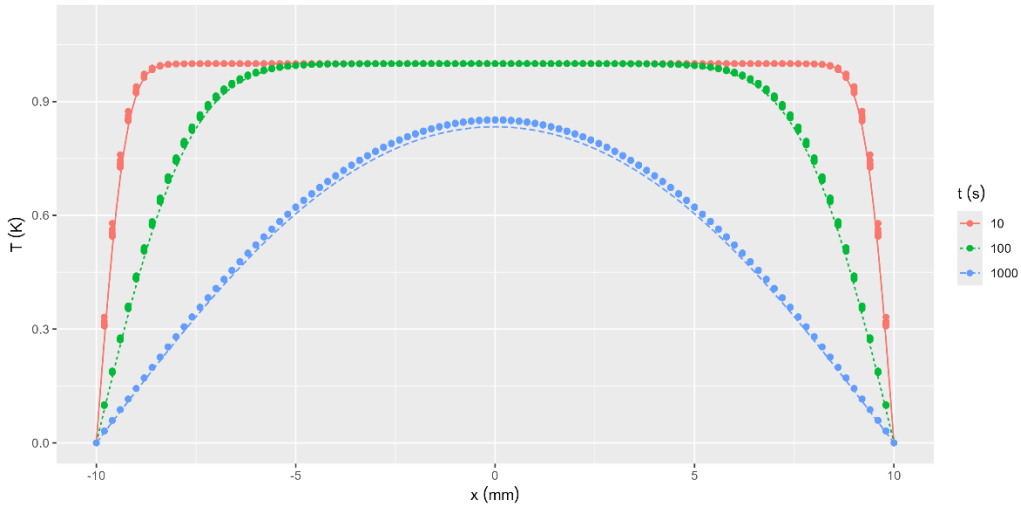

Figure 2: Temperature profile predictions compared with analytical solution for the 1D heat flow test for particle spacing $dp = 0.2$.

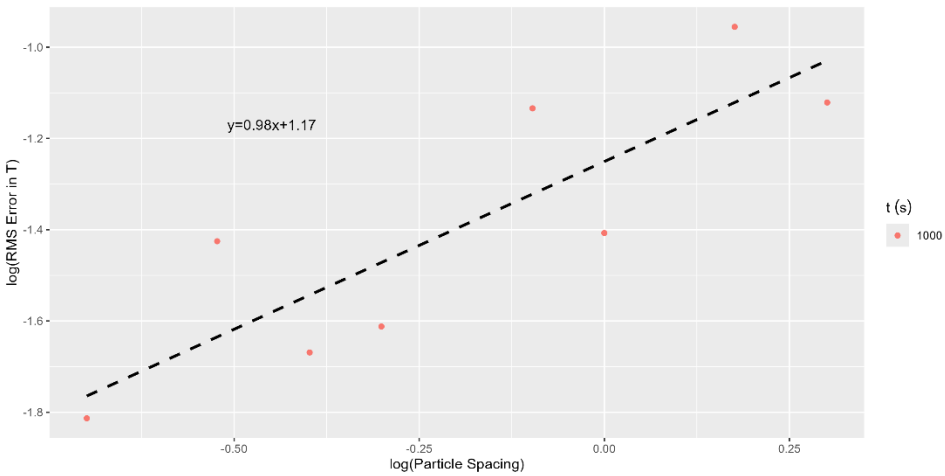

Figure 3: Order of convergence estimate for the error in temperature profile for t=1000 for the 1D heat flow test.

5.1.2 1D Heat generation

Found under: Verification Cases/1D Transient Heat Source

The case assumes a 1D material with length $L = 20$mm and constant temperature zones at both ends ($x < -10, x > 10$) which are set to a temperature $T = 0K$. The initial system temperature is set to $T_0 = 0K$. The heat generation for the system is set to $\dot{q} =$ 1W/g. The analytical solution for the evolution of temperature with time is given by:

$$T(x,t) = \frac{\dot{q}}{\kappa}\left[\frac{L^2}{8} - \frac{X^2}{2} + \sum_m \frac{-1^m 4L^2}{\pi^3(2m-1)^3}\cos\left(\frac{(2m-1)\pi x}{L}\right)e^{-\left(\frac{(2m-1)^2\pi^2\kappa t}{\rho c_p L^2}\right)}\right] \quad (27)$$

Where κ is the thermal conductivity, c_p is the specific heat and ρ is the density.

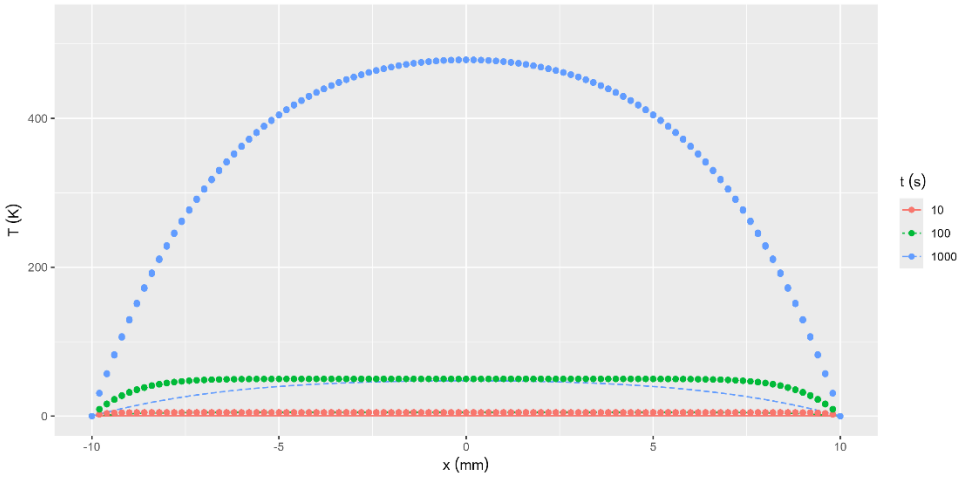

Figure 4: Temperature profile predictions compared with analytical solution for the 1D heat source test for particle spacing $dp = 0.2$.

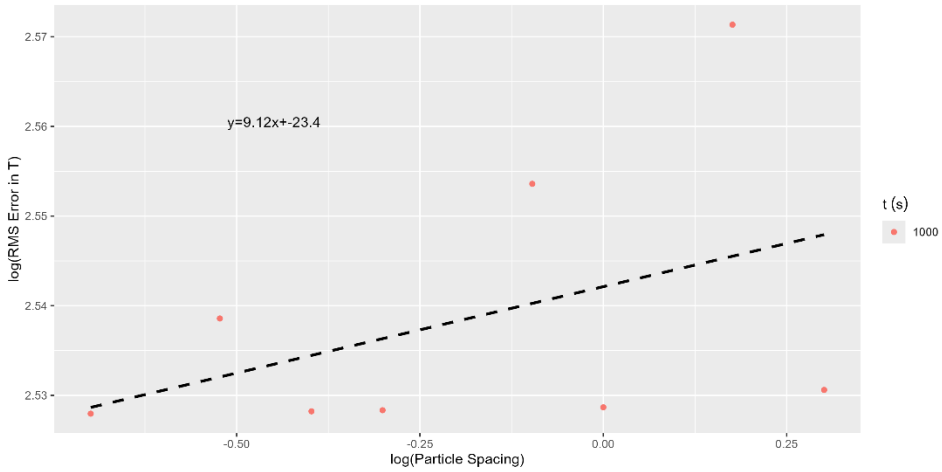

Figure 5: Order of convergence estimate for the error in temperature profile for t=1000 for the 1D heat source test.

5.1.3 1D Material Interfaces

Found under: Verification Cases/1D Material Thermal Interface

The case assumes a 1D material with length $L = 10$mm and constant temperature zones at both ends ($x < 0, x > 10$) which are set to a temperature of $T_C = 0K$ and $T_H = 1K$ respectively. The initial system temperature is set to $T_0 = 0K$. The thermal conductivity for the system is set to $\kappa_1 = 1$, $\kappa_2 = 10$ for $x < 5$ and $x > 5$ respectively. The analytical solution for the steady state temperature is given by:

$$T(x) = \begin{cases} (T_H - T_C)\dfrac{2\kappa_2\left(x + \frac{L}{2}\right)}{L(\kappa_1 + \kappa_2)} + T_C & x < L/2 \\ (T_H - T_C)\dfrac{1 + 2\kappa_1\left(x - \frac{L}{2}\right)}{L(\kappa_1 + \kappa_2)} + T_C & x > L/2 \end{cases} \tag{28}$$

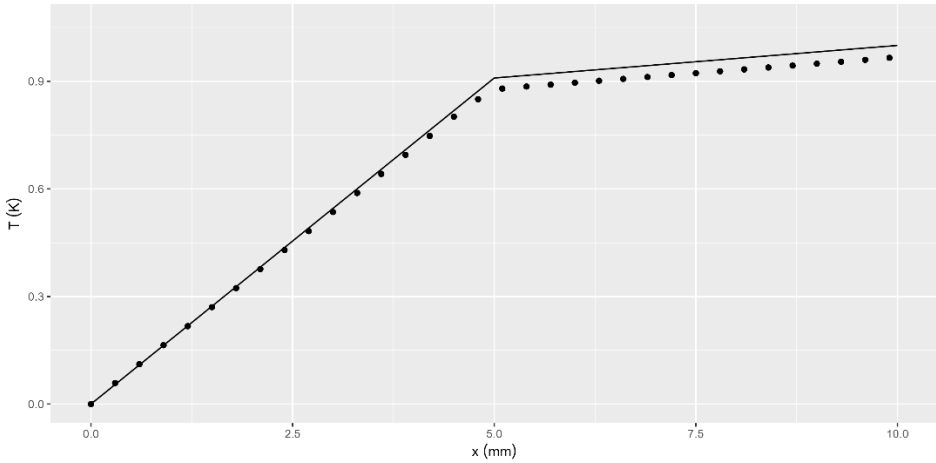

Figure 6 Temperature profile predictions compared with analytical solution for the 1D material interface test for particle spacing $dp = 0.3$.

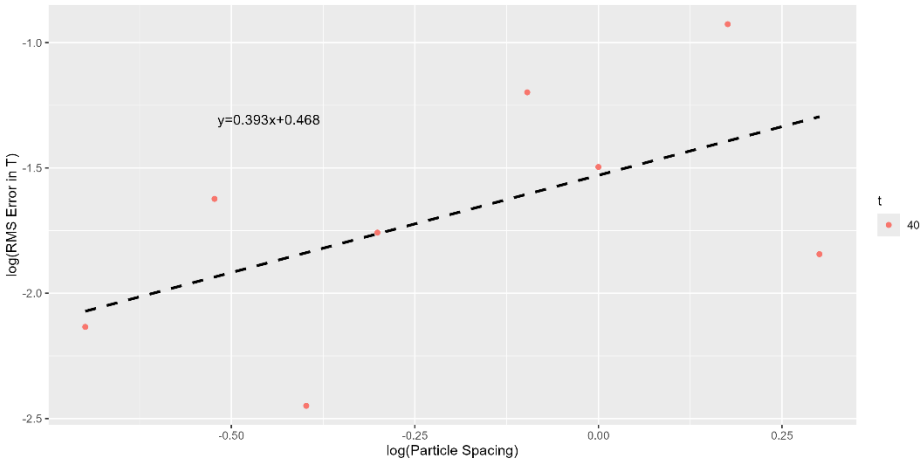

Figure 7: Order of convergence estimate for the error in temperature profile for t=40 for the 1D material interface test.

5.1.4 1D Convection

Found under: Verification Cases/1D Transient Convection

The case assumes a 1D material with length $L = 20$mm and constant temperature zones at one end $x < 0$ which is set to a temperature $T = 0K$. The other end $x > 20$ is set to have a convective boundary based on newtons law of cooling with a coefficient of convection $h = 20$. The initial system temperature is set to a linear profile based on position $T_0 = x$. The analytical solution for the evolution of temperature with time is given by:

$$T(x,t) = \sum_n a_n \sin(\lambda_n x)\, e^{-\left(\frac{\lambda_n^2 \kappa t}{\rho c_p}\right)} \tag{29}$$

$$a_n = \frac{2(\sin \lambda_n L - \lambda_n L \cos \lambda_n L)}{\lambda_n(\lambda_n L - \sin \lambda_n L \cos \lambda_n L)} \tag{30}$$

Where λ_n is calculated as the roots of the equation $\tan \lambda L = -\kappa\lambda/h$.

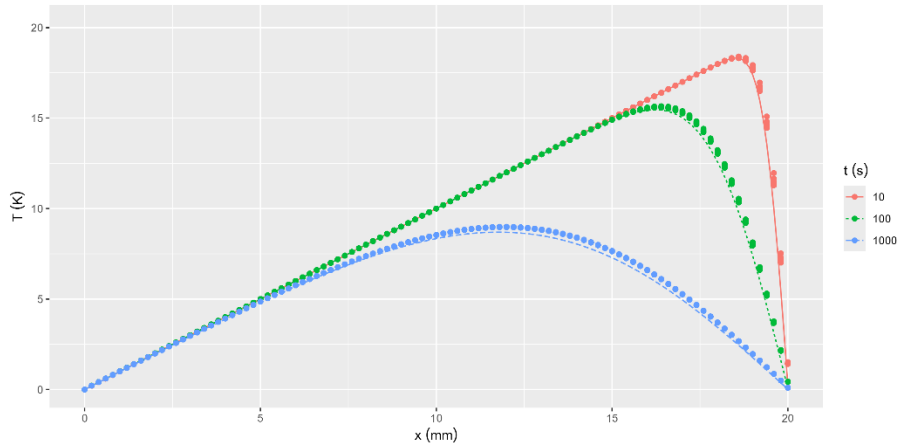

Figure 8 Temperature profile predictions compared with analytical solution for the 1D convection test for particle spacing $dp = 0.2$.

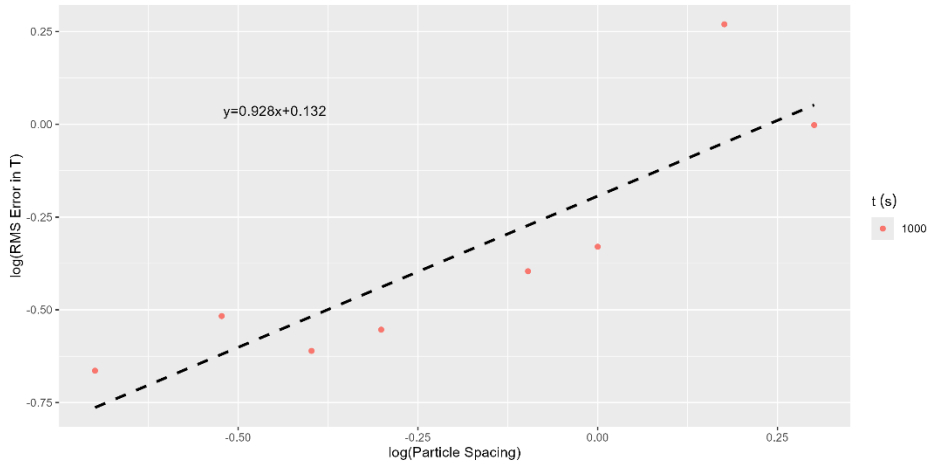

Figure 9: Order of convergence estimate for the error in temperature profile for t=1000 for the 1D convection test.

5.1.5 1D Fuel & Cladding Model

Found under:

5.1.6 2D Fuel and Cladding Model

Found under:

5.1.6.1 Centre Line Temperature Predictions

Found under: Verification Cases/Centreline Convection

This variation on the case includes comparison data taken from the open literature [8] for ENIGMA.

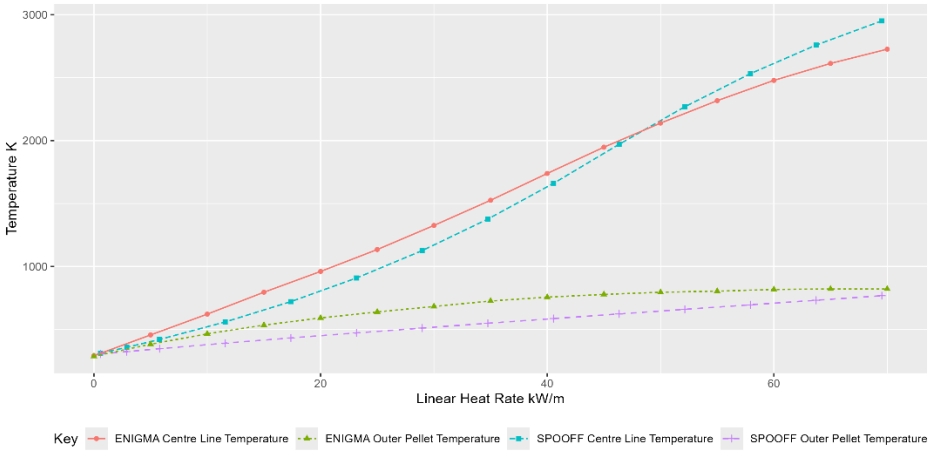

Figure 10: Comparison of centre line temperature predictions with ENIGMA predictions taken from the literature [8].

5.2 Mechanical

5.2.1 Elastic Tensile Test

Found under:

5.2.2 Plastic Tensile Test

Found under:

5.3 Thermo-Mechanical

5.3.1 1D Thermal Induced Strain

Found under: Verification Cases/1D Thermal Induced Stress

The case assumes a 1D material with length $L = 20$mm and constant temperature zones at both ends ($x < -10, x > 10$) which are set to a temperatures $T = 0K$ and $T = 10K$ respectively. The initial system temperature is set to $T = 0K$. The system velocities are fixed which prevent the system from moving and therefore allows a check of the instantaneous stress produced by heating before expansion occurs. The linear thermal expansion coefficient for the system is set to $\alpha = 0.001$ and the young's modulus $E = 100$. The analytical solution for the stress σ is given by:

$$\sigma(x) = -\alpha E\left[(T_H - T_C)\frac{x}{L} + (T_H + T_C)\frac{1}{2}\right] \tag{31}$$

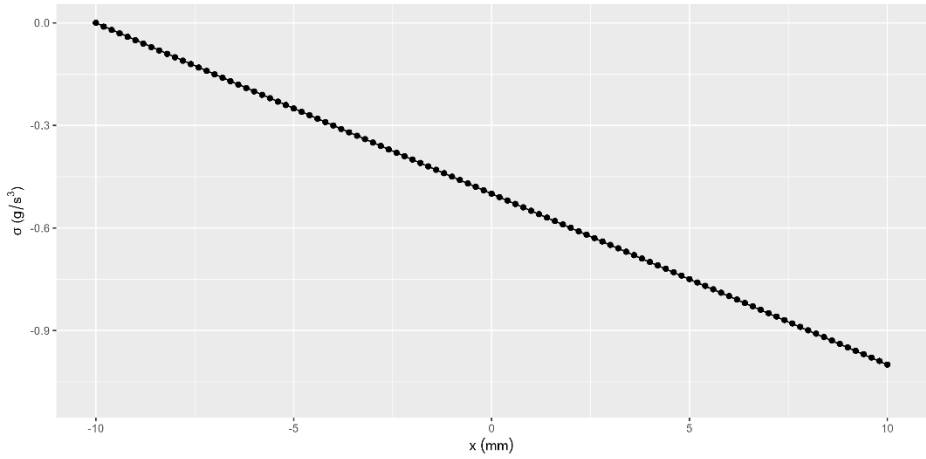

Figure 11: Temperature profile predictions compared with analytical solution for the 1D thermal stress test for particle spacing $dp = 0.2$.

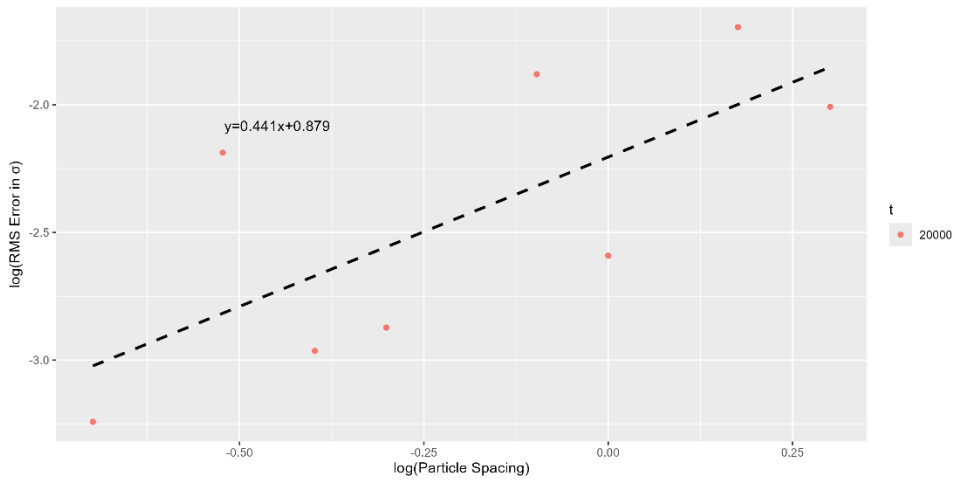

Figure 12: Order of convergence estimate for the error in temperature profile for t=1000 for the 1D thermal stress test.

5.3.2 2D Thermal Induced strain in an annulus

Found under: Verification Cases/2D Annular Thermal Induced Stress

(Just thermal case under: Verification Cases/2D Annular Heatflow)

The case assumes a 2D annular material with inner radius $A = 1$mm, outer radius $B = 12$mm and constant temperature zones at both ends ($r < 1.5, r > 11.5$) which are set to at temperatures $T_i = 1K$ and $T_o = 0K$ respectively. The initial system temperature is set to $T = 0K$. The system is free to expand un constrained. The linear thermal expansion coefficient for the system is set to $\alpha = 0.001$, the young's modulus $E = 100$ and the

Poisson's ratio $\nu = 0.3$. The analytical solution for the steady state temperature and radial stress σ_r are given by:

$$T(x) = \frac{T_i \ln\frac{B}{r} + T_o \ln\frac{r}{A}}{\ln\frac{B}{A}} \tag{32}$$

$$\sigma_r(x) = \frac{E\alpha}{(1-\nu)\ln\frac{B}{A}}\left[\frac{T_o}{2}\left(\ln\frac{r}{A} + B^2\left[1-\frac{A^2}{r^2}\right]\frac{\ln\frac{B}{A}}{B^2-A^2}\right) - \frac{T_i}{2}\left(\ln\frac{B}{r} + A^2\left[1-\frac{B^2}{r^2}\right]\frac{\ln\frac{B}{A}}{B^2-A^2}\right)\right] \tag{33}$$

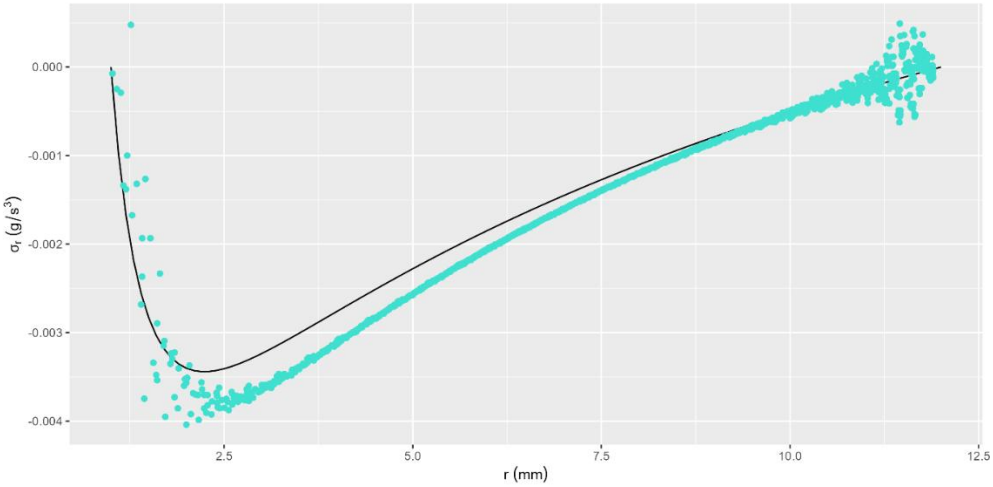

Figure 13: Radial stress profile predictions compared with analytical solution for the 2D thermal stress test for particle spacing $dp = 0.2$.

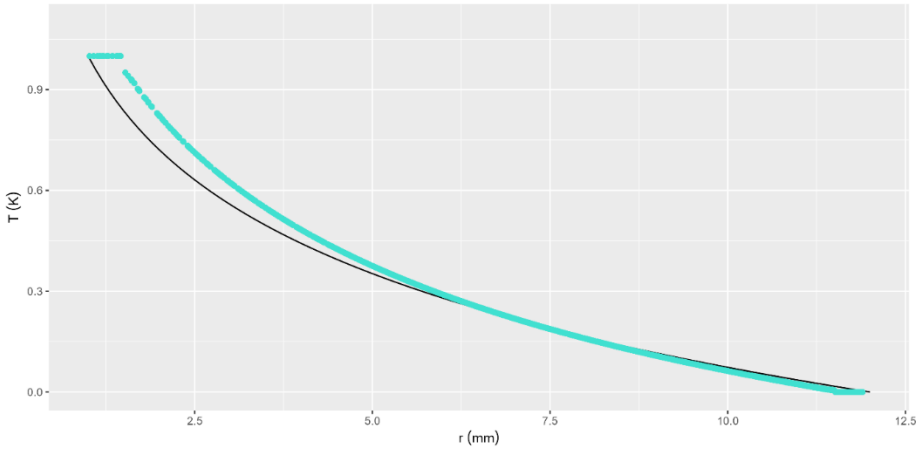

Figure 14: Temperature profile predictions compared with analytical solution for the 2D thermal stress test for particle spacing $dp = 0.2$.

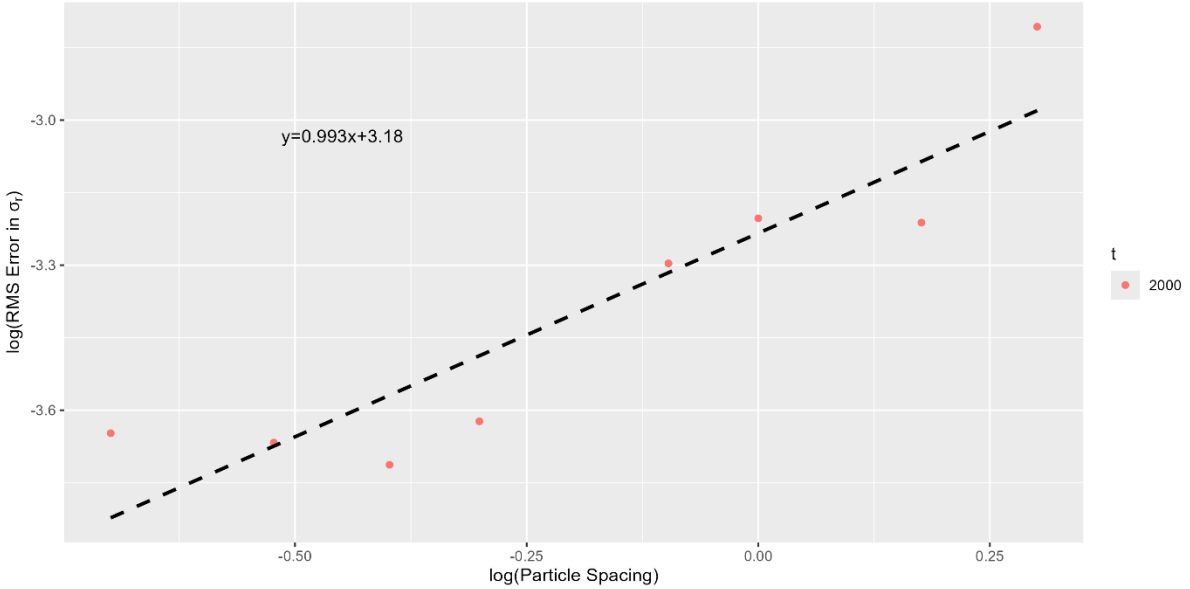

Figure 15: Order of convergence estimate for the error in radial stress profile for t=2000 for the 2D thermal stress test.

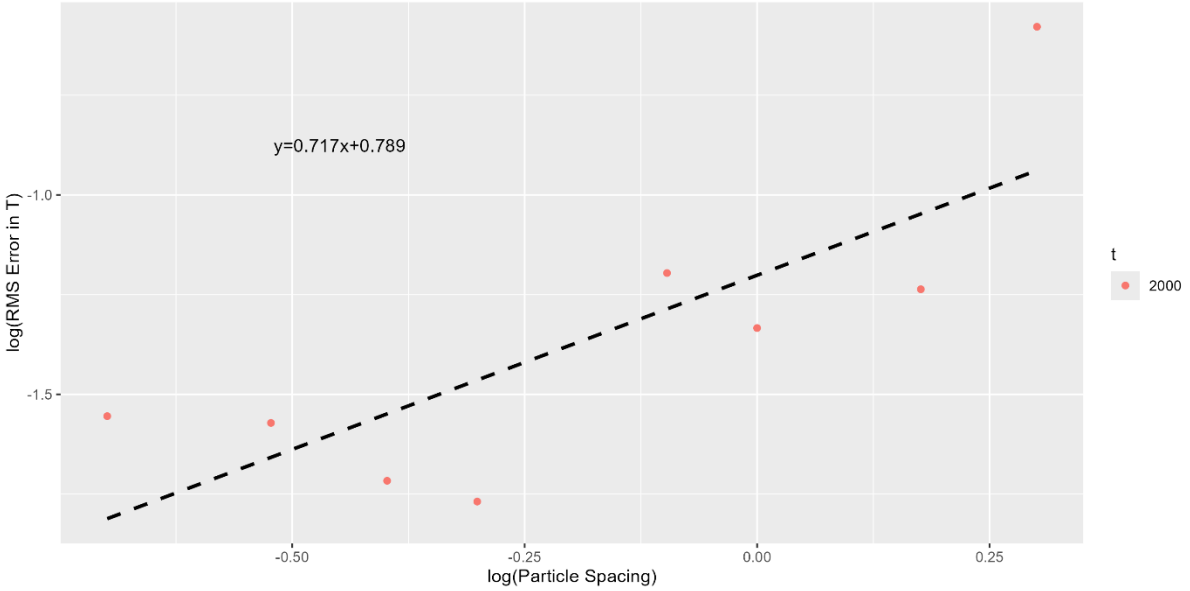

Figure 16: Order of convergence estimate for the error in temperature profile for t=2000 for the 2D thermal stress test.

5.4 Fracture

Found under: Verification Cases/Cracks

This test case demonstrates the ability of the model to calculate material failure for problems with heat generation and thermal induced stresses. The fuel is divided into 2 regions for the body of the fuel and the fixed temperature boundary of the fuel. The fuel pellet radius is assumed to be 4.1mm. The fuel outer boundary is assumed constant at 500K. A particle mesh is generated with a spacing of 0.1mm is used. The fuel region is assumed to be UO_2 with temperature dependant property models for specific heat, thermal conductivity, elastic modulus and fracture strength.

12 cases of linear heat rate are tested. Further comparison of data for the test case are given in OperaHPC Deliverable D4.3 [3]

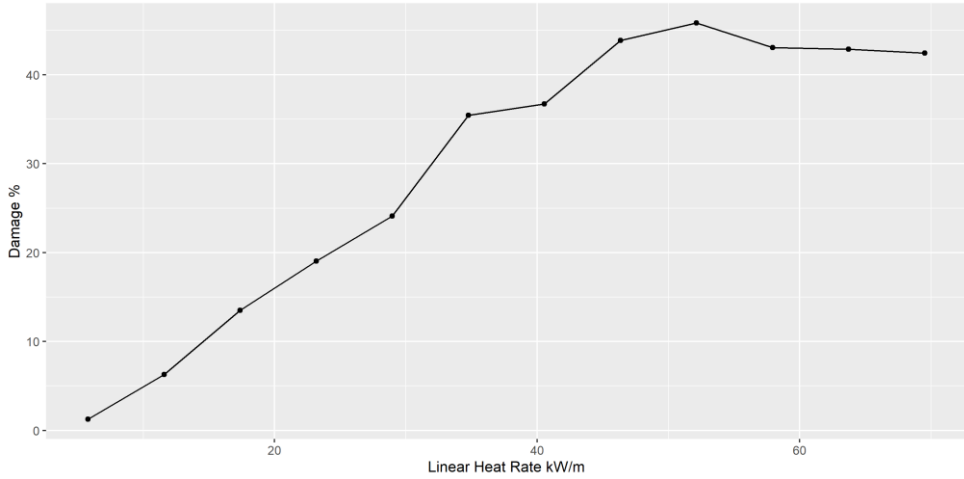

Figure 17: Average per particle damage at time step 390 across all linear heat rates.

5.5 Pellet Cladding Mechanical Interaction

Found under: Verification Cases/Pellet Cladding Mechanical Interaction

This test case demonstrates the ability of the model to calculate interaction of moving fragmented material. The material is considered as 3 section - fuel, cladding and gap – with only the fuel and cladding regions containing interpolation points.

The heat generation of the fuel is considered around a value of roughly 70kW/m, heat generation is therefore set to 120W/g in line with the maximum fragmentation observed in the fracture case.

The fuel region is assumed to be UO_2 with temperature dependant property models for specific heat, thermal conductivity, elastic modulus and fracture strength and with a constant linear expansion coefficient of $1\text{x}10^{-5}$. The cladding region is modelled as plastic with a yield strength of $400MPa$ and temperature dependant model for fracture strength. The cladding is modelled with a linear expansion coefficient of $6\text{x}10^{-6}$.

Contact between the cladding and fuel as well as the fuel-to-fuel fragments and cladding to cladding fragments is modelled using the damped Hertz contact force. The force scaling is set to 1 for the fuel to cladding interaction and to 0.5 for the self-contact forms. For all forms the damping scaling is set to 1. To increase stability and limit interpenetration of material the integrator is set to limit velocities to 30m/s.

A post processing script able to estimate fragment numbers and sizes based on bon analysis is included with the case.

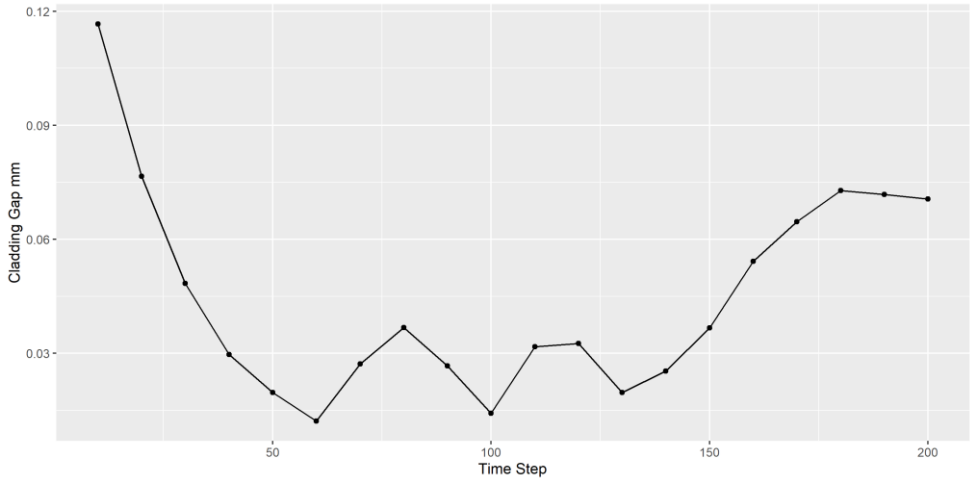

Figure 18: Minimum distance between cladding and fuel interpolation points over time for the for the 70kW/m case.

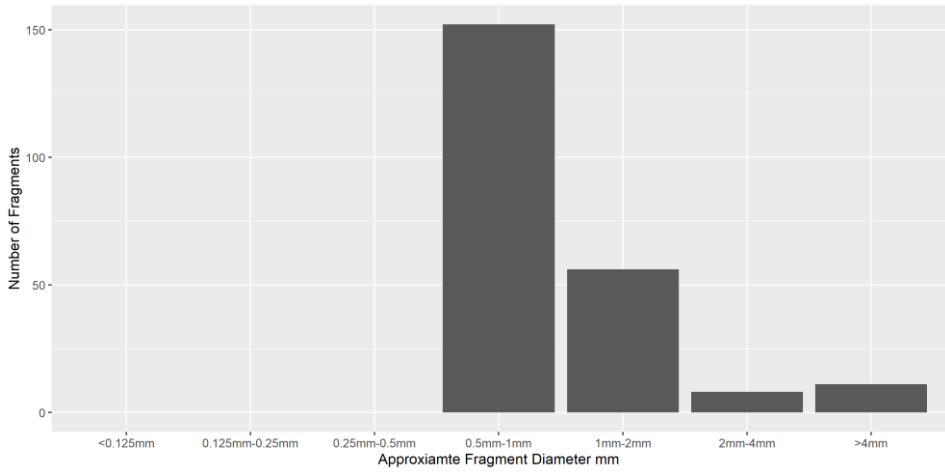

Figure 19: Number of fragments calculated by R script with Damage threshold set to 0.1 for the 70kW/m case. Fragments are approximated as circles for diameter estimates.

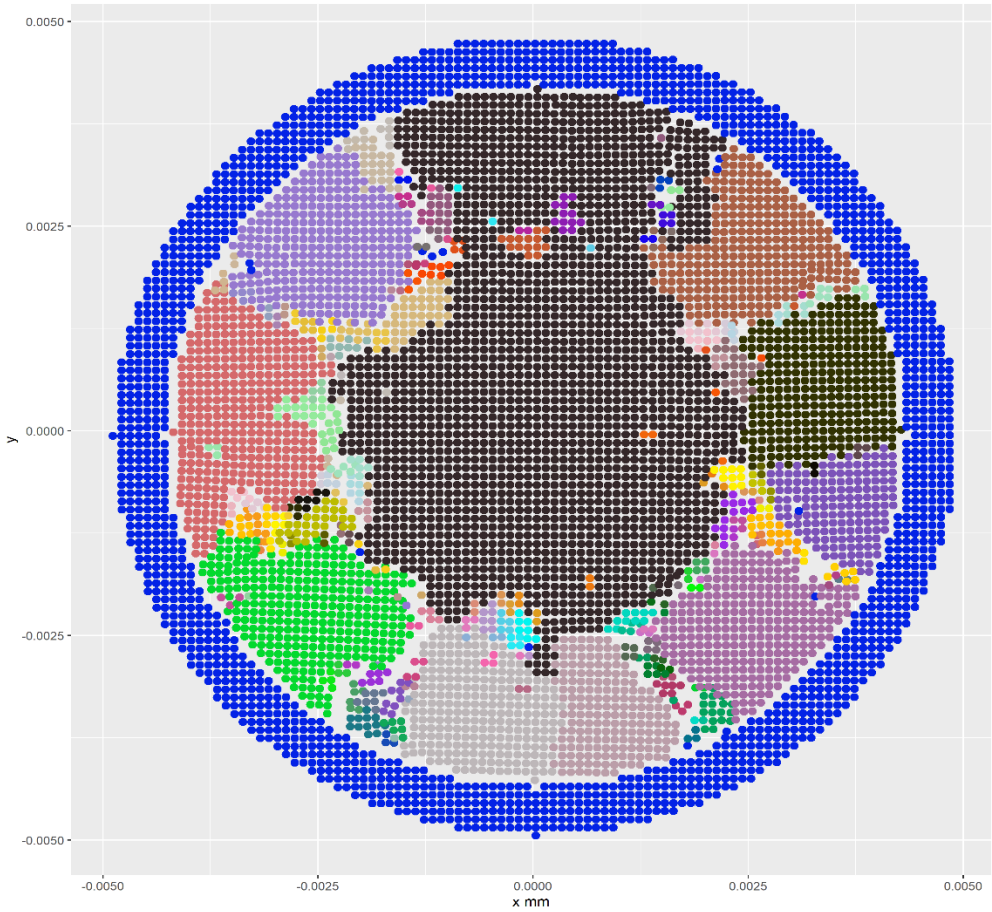

Figure 20: Fragment map generated by R script with Damage threshold set to 0.1for the 70kW/m case

6 References

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